When Does Filtering Help You See a Break? Latent-State Diagnostics for Structural Change at Calibrated False-Alarm Rates

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Abstract

A two-layer framework for detecting hidden structural change: a state-estimation layer (Kalman filtering of a latent state from noisy observations) and a diagnostics layer that converts features of the strictly-causal filtered path into alarms. All detectors — ours and benchmarks — are calibrated on matched null data-generating processes to the same false-alarm rate (5% per 500 observations), making detection rates and delays directly comparable. The main finding is a division of labor, not a victory: for level shifts, a CUSUM on the raw data dominates detection rates at every signal-to-noise ratio, while the latent-innovation CUSUM is "fast or never" — a phenomenon we formalize (the post-break innovation mean decays geometrically to $\delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$; when this is below the CUSUM allowance, post-transient detection has an exponentially small bound) and verify numerically. For second-moment and dynamics changes the ordering reverses — though a whitening ladder (the same variance CUSUM on raw data, ARIMA residuals, and Kalman innovations) shows the edge is *prewhitening under autocorrelation*, not the latent state estimate itself: a raw-data variance CUSUM given identical calibration detects the subtle (×1.5) break only when observation noise dominates (0.996 at SNR 0.1) and falls to chance as the latent signal grows (0.560, 0.102 at SNR 0.5, 2.0), whereas the same statistic on prewhitened residuals detects it at every SNR (ARIMA 0.90/0.94/0.87, tracking the latent composite's 0.82/0.87/0.91). The diagnostics further survive heavy tails via an exceedance-indicator variant and, via a shortfall CUSUM, detect variance *quieting* that level-oriented methods miss. A real-data application (industrial production, GDP, Treasury yields) reproduces the profile: every alarm attributes to a second-moment feature, association with NBER reference dates is significant (permutation p = 0.008), and real-time (ALFRED vintage) analysis confirms the COVID timing while honestly downgrading the 2008 timing claim.

JEL classification: C12, C22, C52. Keywords: structural change; sequential change detection; CUSUM; state-space models; Kalman filter; prewhitening; false-alarm rate; latent-state diagnostics; volatility regimes.

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1. Introduction

Structural change is usually sought in observables. But many economic quantities of interest — trend output, a natural rate, an underlying volatility regime — are latent states observed only with noise, and the natural first step is to filter: estimate the state, then watch the estimate for breaks. Intuition says the filtering should help, by stripping observation noise before the change is sought. This paper asks when that intuition holds, under a protocol that makes the question answerable. Our contributions are as follows.

- Contribution 1 (protocol): a calibrated-FAR parity harness. Every method's threshold comes from the same routine on the same matched-null draws; empirical FAR is re-verified on fresh nulls; causality is enforced structurally (parameters fit on a training prefix only; forward-only filtering; alarm scores are NaN on training data) and tested bit-identically (perturbing future observations leaves all scores at earlier times unchanged, exactly).
- Contribution 2 (negative + positive results): the intuition is wrong for first moments at every SNR — and we can prove why (fast-or-never theorem). For second moments it is right, but a whitening ladder (raw / ARIMA / Kalman, same variance CUSUM) localizes *why*: the advantage over raw data is prewhitening under autocorrelation — decisive at high SNR, where the latent signal masks the observation-noise change and the raw detector sits at chance — and the latent state estimate itself adds little once the observations are whitened (ARIMA residuals suffice).
- Contribution 3 (method): a tail-robust exceedance-indicator CUSUM that preserves the second-moment advantage under heavy-tailed noise, found via two documented failed designs.
- Contribution 4 (application discipline): attribution, permutation tests, sensitivity, pinned data snapshots, and real-time vintages for the real-data claims — including a self-correction on the headline GFC timing.
- Honest-outcome framing throughout: three pre-registered hypotheses were falsified; every post-hoc change is logged (CHANGELOG) and the failures are reported as findings.

Related work. The paper sits at the intersection of four literatures. Sequential change detection descends from Page's (1954) CUSUM and the quickest-detection tradition (Lorden 1971; Moustakides 1986); we use the CUSUM as the common statistic across information sets rather than proposing a new stopping rule. Regime-switching models (Hamilton 1989; Kim–Nelson 1999) offer a state-aware alternative whose regime probabilities we include as a benchmark and find saturate under calibration on nonstationary data. The empirical target of the second-moment results is the Great Moderation volatility decline (McConnell–Perez-Quiros 2000; Stock–Watson 2002), which motivates the variance and quieting scenarios. Finally, offline changepoint methods (PELT, Killick et al. 2012; and the wider changepoint literature) solve a retrospective segmentation problem; our monitoring is strictly causal and calibrated to a false-alarm rate, so the comparison is to online detectors on matched nulls. What is new here is not any single detector but the calibrated-parity harness that makes latent-state and raw-data detectors directly comparable, and the whitening ladder that decomposes the latent layer's second-moment advantage into a prewhitening component and a (negligible) state-estimation component.

2. Framework and evaluation protocol

Model layer. $S_t = \varphi S_{t-1} + w_t$ (var q), $Y_t = S_t + v_t$ (var r); the estimator sees only Y and fits (φ, q, r) by maximum likelihood on a training prefix (25% of the sample), then runs a forward-only filter with frozen parameters. Standardized one-step innovations e_t and the filtered state are the raw material for diagnostics.

Diagnostics layer. Eleven features of the filtered path — level change, slope, acceleration, instability, rolling persistence, a Page CUSUM of innovations (allowance $k = 0.5$), variance CUSUMs of e^2-1 ($k = 0.25$ and 0.05), a quietness CUSUM of $1-e^2$ ($k = 0.05$), rolling innovation autocorrelation, and a CUSUM of the filtered state against its training baseline. Each feature is standardized at every time point by the median and IQR of its null distribution at that same time point (pooled-over-time standardization is a design flaw that blunted the composite; §8.4). The composite score is the max of standardized features; the composite is itself calibrated on nulls, which prices in the multiple-feature testing automatically.

Calibration parity. Threshold = (1−FAR) quantile of the per-replication maximum score over matched-null draws; identical routine, budgets, and seed layout for LSC detectors and benchmarks (raw-Y CUSUM, ARIMA+CUSUM, plain-HMM regime flips). Calibration, evaluation, FAR-check, and feature-scale seeds are disjoint by construction.

Metrics. Detection rate, pre-break false alarms, censored delay; for multi-break: event-level precision/recall/F1 with one-to-one greedy matching in a 100-observation window.

3. Simulation design

Arenas: AR(1) latent state ($\varphi = 0.95$) at spec-SNR (stationary state variance / observation variance) $\in \{0.1, 0.5, 2.0\}$; a local-level (random-walk state) arena is retained as a documented degenerate case (§8.4). Breaks at mid-sample: level shifts (0.5, 1, 3 σ_ref), logistic ramps, observation-noise scale changes (×1.5, ×3, and ×⅔ quieting), and pure persistence changes ($\varphi \to 0.995$ or 0.80 with the stationary variance held fixed). T = 500 baseline with a $T \in \{200, 2000\}$ sweep; misspecification arenas with t_5 observation noise and a nonlinear tanh state drift.

4. First moments: raw data wins, and we can say exactly why

Empirics. At matched FAR, raw-Y CUSUM has the best level-shift detection rate at every SNR (0.97–0.99 at 3σ) and every T; the latent state CUSUM approaches but never overtakes it (0.19 $\to$ 0.94 as SNR rises). Two pre-registered "latent advantage" hypotheses were falsified. The latent-innovation CUSUM is the speed champion conditional on firing: median delay 24–53 observations at 3σ versus raw's 58–91, at the cost of detection rate (0.55–0.67) — "fast or never."

Theory. The empirics above are not an artifact of tuning; they follow from two facts about the detectors' post-break drift. Derivations are in Appendix B (`experiments/THEORY.md`).

> Proposition 1 (fast-or-never). After a state level shift δ, the standardized innovation mean decays geometrically at rate $\rho = \varphi(1-K)$ to $\mu\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$, where K is the steady-state Kalman gain and F the innovation variance. If $\mu\infty < k$, the post-transient innovation CUSUM has negative drift, and the probability of an alarm in the next L observations is bounded by $(L+1)\cdot\exp(-2(k-\mu\infty)h)$ for threshold h.

The detector therefore fires during the adaptation transient or, with exponentially small probability, never. With $\varphi = 0.95$ and $k = 0.5$, $\mu\infty > k$ would require shifts of order 10σ: the innovation CUSUM is *structurally* in the fast-or-never regime, which is why it is the speed champion but never the detection champion.

> Proposition 2 (raw-CUSUM delay). The raw-Y CUSUM sees the full shift as a sustained standardized drift Δ = δ/σ_Y, giving an Albert–Wald mean delay of approximately $h/(\Delta-k)$ once $\Delta > k$, and negligible power when $\Delta \leq k$.

Verification (`experiments/exp06_theory_check.py`, 1000 reps): the Monte Carlo innovation path matches μ_t within MC error; the Proposition 1 bound is never violated; at the actual calibrated thresholds, $\mu\infty$ sorts every observed detection rate — $\delta \leq 1\sigma$ gives bound $\leq 0.7\%$ (observed $\approx$ FAR); $\delta = 3\sigma$ is a knife-edge ($\mu\infty = 0.43$–0.48 vs $k = 0.5$) matching the observed 0.55–0.67. Proposition 2's Wald delays 68/84/110 across SNRs bracket the observed raw medians 58/75/91 ($\approx$15–20% conservative) and explain the partial 1σ detection (0.30) without any fitting: $\Delta = 0.577$ barely exceeds k, so the Wald delay (1334) dwarfs the 250-observation horizon.

5. Second moments: a whitening ladder

Is the latent layer's second-moment advantage about the *state estimate*, or merely about *prewhitening* the autocorrelated observations before applying a variance statistic? We answer this directly by running the identical variance CUSUM (up-arm Page CUSUMs of z^2-1 at allowances $k = 0.25$ and 0.05, a down-arm CUSUM of $1-z^2$, max over arms, no per-time standardization) at three levels of prewhitening, calibrated by the same routine on the same matched-null seed blocks:

- raw — z from the raw observations, standardized by frozen training-prefix moments;
- ARIMA — the same statistic on the standardized one-step residuals of an AIC-selected, training-prefix-frozen ARIMA model (whitened, but not state-aware);
- latent — the composite's e^2-based variance features on the Kalman innovations (state-aware).

The ladder (detection rate at $T = 500$, 5% calibrated FAR; MC SEs ≤ 0.02 in `paper_assets/ladder_table.csv`).

break	rung	SNR 0.1	SNR 0.5	SNR 2.0
variance ×1.5	raw	1.00	0.56	0.10
	ARIMA	0.90	0.94	0.87
	latent	0.82	0.87	0.91
variance ×3	raw	1.00	1.00	0.85
	ARIMA	0.98	1.00	1.00
	latent	0.99	0.99	0.98

Reading the ladder. The subtle ×1.5 break is the discriminating case. The *raw* rung is strongly SNR-dependent — it falls monotonically from 0.996 (SNR 0.1) through 0.560 (SNR 0.5) to 0.102 (SNR 2.0, within 5 pp of the 6.0% empirical FAR, i.e. chance). The mechanism is transparent: as SNR rises the latent state's variance dominates the marginal variance of Y, so a ×1.5 change in the (now-small) observation-noise component is a shrinking fraction of total variance and is masked by state-driven autocorrelation. This overturns the scoped claim — earlier framed against "level-oriented detectors standard in this literature" — that a raw-data variance detector must sit at chance: it does not, *when observation noise dominates*. But prewhitening removes exactly the autocorrelation that masks the break: the *ARIMA* rung is essentially flat across SNR (0.90 / 0.94 / 0.87) and tracks the *latent* rung (0.82 / 0.87 / 0.91) step for step — the state estimate adds little the ARIMA residuals do not already provide. The advantage over raw is therefore *prewhitening under autocorrelation*, not latency or state estimation per se (the recipe in §10 is revised accordingly, and the fast-or-never first-moment result of §4 is untouched — that concerns the state CUSUM, not the variance CUSUM). At the coarser ×3 break every rung is at or near ceiling; the ladder only separates in the subtle regime. This is the pre-registered Outcome C of the decision rule logged in `experiments/CHANGELOG.md` (Outcomes A and B, which required the raw rung to be uniformly at chance or uniformly within 10 pp of the composite, are both falsified by the SNR dependence). Attribution: the z^2/e^2 variance-pressure arms drive the alarms at every rung; on real data the same features drive every crisis alarm (§9).

Runway (T sweep, ×1.5, SNR 0.5). The subtle-break detection scales with sample length for every rung: at $T = 200$ all three are weak (raw 0.26, ARIMA 0.10, latent composite 0.11) — the whitened rungs need runway to accumulate the CUSUM — and by T = 2000 all reach ceiling (raw 0.98, ARIMA $\ap-prox$1.00, latent 0.99). The T = 200 calibration is *not* hot for these variance detectors (empirical FAR 5.0–5.8% vs the 5% target), unlike the level CUSUMs (§8.1). At the coarse $\times 3$ break the composite is at ceiling even at T = 200 (1.00, median delay 26 of the ~100 post-break observations, best level-oriented benchmark 0.17).

Scale *quieting* (×⅔) inverts the up-break pattern and is treated in §6 and §8.3: there the raw rung catches the low-SNR case (0.32 at SNR 0.1, where noise dominates) while the whitened rungs and composite sit at chance, and only the standalone exceedance detector recovers it at moderate SNR.

6. Dynamics: near the information floor

Pure persistence changes (marginals preserved) are close to undetectable: at SNR 0.5, no method exceeds FAR (a pre-registered hypothesis wrong in both directions — raw CUSUM even reached 0.16 on persistence-up via a conditional "level-freeze" artifact we dissect). Quieting changes (φ down) actively *suppress* every excursion statistic below its null level. Purpose-built quietness features (CUSUM of $1-e^2$, rolling innovation autocorrelation) rescue detection only where the information exists: 0.33 at SNR 2.0 (the only above-FAR persistence detection anywhere in the grid) and 0.17 at T = 2000. Scale-quieting (×⅔) is detectable — but only by the exceedance detector (§8.3): 0.41/0.33, all other methods at chance.

7. Multiple breaks: everyone is one-shot

Under a re-arm protocol applied identically to all methods (re-arm when the score drains below half threshold plus a 20-obs refractory), raw CUSUM never detects a second event (recall 0.00 — its fixed-baseline statistic saturates and cannot drain), but the LSC CUSUMs rarely re-arm in time either at 150-observation spacing (level→level second-event recall $\leq$ 0.05): first-alarm delay plus drain time exceeds the gap. The exception is cross-channel pairs: level→variance is caught by the composite alone (second-event recall 0.60, F1 0.63) because its variance features were never saturated by the level event. Re-arming costs almost nothing under the null ($\leq$ 1.2% of null paths give a second alarm) — saturation, not chatter, binds. Bounded-memory (windowed) statistics are the identified fix and are left to future work.

8. Robustness

8.1 Sample size. §5 numbers; short-T caveat: heavy-tailed null maxima make quantile thresholds noisy, and at T = 200 the two baseline CUSUMs calibrate hot (8.8–9.4% vs the 5% target). Quote empirical FARs alongside any short-sample power claim.

8.2 Misspecification. t_5 observation noise: rankings preserved; raw CUSUM unhurt on levels (0.99); the composite's subtle-variance case collapses (×1.5: 0.87 → 0.16) — repaired in §8.3. The raw and ARIMA variance rungs are far less tail-fragile than the composite: on the ×1.5 break at SNR 0.5 the raw rung falls only 0.56 → 0.43 and the ARIMA rung 0.94 → 0.74 under t_5 (against the composite's 0.87 → 0.16), because the plain max-over-arms z^2/e^2 statistic without per-time composite standardization retains the tail excursions that carry the variance signal — the same logic that motivates the standalone exceedance detector (§8.3). At ×3 all variance rungs stay at ceiling under t_5 (raw and ARIMA 1.00, composite 0.97). Nonlinear tanh drift (mildly bimodal state): level detection collapses for *every* method (raw 0.15 at 3σ

— the state's own regime-hopping inflates all null thresholds), while variance detection is untouched (0.97 at ×1.5). The two detection families read disjoint information channels.

8.3 Heavy tails and the exceedance repair (a three-act story worth telling honestly). (i) Huberizing e^2 (clip at 2.5·MAD) — falsified, worse everywhere, even Gaussian (×1.5: 0.87 → 0.06): the variance signal lives in the tail the clip removes. (ii) An exceedance-indicator CUSUM (count of |e| above its training 90th percentile; bounded summand under any distribution) — the raw statistic separates near-perfectly, but dies inside the composite: a max-over-features composite rewards break-to-null-IQR ratio, and bounded increments cannot reach the ratios unbounded features set the threshold with. (iii) The same statistic as a *standalone* calibrated detector (up-arm k = 0.05, down-arm k = 0.02; k chosen on non-evaluation seeds, procedure logged): variance ×1.5 at 0.87 Gaussian / 0.75 t_5 (repairing 0.16), ×3 at ~1.0 with ~37-obs delay under both distributions, and the first successful quieting detection (×⅔: 0.41/0.33). Both headline rates fell 3–5pp short of the pre-registered bars — reported as such.

8.4 Protocol lessons (each cost us a wrong result before it was fixed). Random-walk-state arenas are degenerate for level-break ranking; plain-HMM regime probabilities saturate and cannot be FAR-calibrated on nonstationary data; probability-scale scores need log-odds; EM needs persistent-initialization restarts; composite features must be standardized per-time-point, not pooled; order-statistic thresholds have Beta(n+1−k, k) noise regardless of distribution, so heavy-tailed detectors need larger calibration budgets.

9. Real data (illustrative)

Three FRED series, pinned snapshots (2026-07-11), rolling causal monitoring (train 120 months / monitor 60), per-segment parametric bootstrap calibration at 5% FAR per window, alarms attributed to the feature that crossed.

Industrial production (INDPRO, 1948–2026). Composite alarms: 2008-09 and 2020-04 (both variance_pressure), 1990-12 (variance_quiet), 1969-08 (variance_quiet; within the false-alarm budget and reported as such). Hits 3/9 NBER peaks within 12 months, 1 stray vs 0.7 expected; permutation p = 0.008 (innovation CUSUM p = 0.018; raw CUSUM 1 hit, p = 0.15). The raw variance CUSUM — the bottom rung of the ladder, added here to test real-data uniqueness — does catch the GFC (2008-09, up-arm, the same month as the composite) but embeds it among four stray quieting alarms (1967, 1968, 1988, 2019) and misses COVID, so its NBER association is not significant (5 alarms, 1 hit, p = 0.56). The crisis *is* detectable by a raw variance statistic; what the latent layer buys on real data is a *clean* alarm profile — a significant, low-stray association — not the crisis catch itself.

What real-time data changes (ALFRED vintages). Re-running each decision on the data *as it existed that month* both tempers and sharpens the timing claims. COVID: robust — the composite's real-time alarm is at the 2020-04 vintage (data month 2020-03), one month earlier than on revised data and ~2 months before the NBER announcement. GFC: downgraded — the crossing is at the same data month (2008-09) but only in the 2008-12 vintage, so real-time knowledge is coincident with the NBER announcement (2008-12-01), though still 4 months ahead of the raw level CUSUM's real-time alarm (2009-04). Because the raw variance CUSUM alarmed on revised INDPRO, we ran it through the identical vintage protocol for an apples-to-apples comparison: its real-time timing is *indistinguishable from the composite's* — same vintage and same data month for both GFC (2008-12 vintage, 2008-09) and COVID (2020-04 vintage, 2020-03), both ahead of the raw level CUSUM. On real-data crisis *timing*, then, the prewhitening rung and the state-aware composite are interchangeable; the composite's advantage is confined to the clean association profile of §9 and to the simulation-calibrated ×1.5 subtlety threshold of §5, not to real-time crisis detection.

GDP (GDPC1, quarterly). GFC and COVID caught (variance_pressure, 2008Q4 and 2020Q2). The raw variance CUSUM again catches both crises (2009Q2 and 2020Q2, up-arm) with fewer alarms than on INDPRO (2 alarms, 1 hit, $p = 0.31$) — the same "catches the crisis, misses the clean association" pattern. The registered Great Moderation event (1984Q1) is an honest miss: causal rolling monitoring detects the quieting only in 1992Q2 (composite) / 1997Q1 (tail shortfall), because 60-quarter training windows contain 1970s volatility until the early 1990s. Retrospective full-sample break dates are not reproducible by honest monitoring.

10-year Treasury yield changes (GS10). Volcker regime caught 4 months after the registered 1979-10 event (variance_pressure; raw and innovation CUSUM the same month); the post-Volcker disinflation quietings are flagged (1989–1996, shortfall/variance_quiet); the 2008 ZLB event is missed within 12 months. The raw variance CUSUM also flags Volcker (1980-02, up-arm) but is the noisiest detector on this series (9 alarms, 8 stray, $p = 0.28$) — on a series that is *all* volatility regime, a raw z^2 statistic fires on every regime shift, which is exactly why its event-association washes out.

Sensitivity. FAR = 10% behaves as expected. Training window 180 months breaks both variance-based detectors' bootstrap calibration on nonstationary real data — the composite and the raw variance CUSUM each fire 14 alarms / 21 windows — while the level (raw CUSUM, 2), innovation (3), and tail (5) detectors stay sane. It is the *second-moment* statistic, not the composite machinery, that is sensitive to a training window long enough to straddle a volatility regime; training windows must be short enough to be locally stationary (120 months worked).

10. Discussion

Four points summarize what the harness bought us.

- The latent-state diagnostics layer is not a better level-shift detector; it is a *different instrument*, reading second moments and dynamics that raw-data detectors miss once the latent signal masks them. The whitening ladder (§5) sharpens the claim: the second-moment advantage over raw data is *prewhitening under autocorrelation*, which ARIMA residuals capture as well as the Kalman state — the state layer's distinctive contribution is dynamics (persistence, quieting) and the fast-or-never speed edge on levels, not variance detection per se. The division of labor is now both measured (grids), laddered (raw / ARIMA / latent), and derived (fast-or-never).
- Practical recipe: run raw CUSUM for levels; for scale changes, *whiten first* (ARIMA residuals or Kalman innovations) and run the variance CUSUM — prewhitening, not the latent state estimate, is what buys detection when the series is autocorrelated and the latent signal is strong, while a raw variance CUSUM suffices only when observation noise dominates (low SNR). Use the exceedance-indicator variant under distributional uncertainty (heavy tails), and the composite for breadth. Calibrate everything on matched nulls at a common FAR and report empirical FARs.
- The calibrated-parity protocol itself is a contribution: it exposed every failure mode above, and it is what makes the negative results informative rather than anecdotal.
- Limitations / future work: bounded-memory statistics for multiple breaks; adaptive composite weighting (breadth tax); switching-SSM (Kim filter) model layer; formalizing the persistence-break mechanisms; a vol-regime reference set for scoring the exceedance detector on real data.

Appendix A. Reproducibility

`make all` regenerates every table and figure from pinned seeds (Python 3.14, statsmodels/hmmlearn; `make fred` / `make realdata` / `make realtime` for the data applications, snapshots under `data/`). 84 tests include bit-identical no-lookahead checks for every feature and detector (including the raw and ARIMA variance rungs and a training-freeze check), DGP ground-truth checks, and calibration-parity checks. All post-hoc design changes and pre-registered hypotheses (including the three falsified ones and the two failed robust-feature designs) are in `experiments/CHANGELOG.md`; full experiment narratives in `experiments/FINDINGS.md`; theory derivations in `experiments/THEORY.md` (Appendix B). The complete source, pinned data snapshots, and seed configuration constitute the replication package, to be posted publicly on acceptance and available from the author on request in the interim.

Appendix B. Theory derivations

Full derivations of Proposition 1 (the geometric decay of the standardized innovation mean to $\mu\infty$ and the resulting fast-or-never alarm bound) and Proposition 2 (the Albert–Wald delay of the raw-Y CUSUM under a sustained standardized drift) are in `experiments/THEORY.md`, with the numerical verification in `experiments/exp06_theory_check.py` (1000 replications). The steady-state Kalman gain K and innovation variance F are the fixed-point solutions of the scalar Riccati recursion for the AR(1) state model of §2; $\mu\infty$, ρ, and the bound follow by taking the post-break innovation as a deterministic geometric transient plus mean-zero noise and applying a one-sided Hoeffding bound to the CUSUM increments.

Appendix C. Summary of key quantities

Claim	Number	Source
Raw CUSUM level 3σ detect, all SNRs	0.97–0.99	grid_v1
Innovation CUSUM delay vs raw, 3σ	24–53 vs 58–91 obs	exp02
Innovation CUSUM detect at 3σ	0.55–0.67	exp02/grid_v1
$\mu\infty$ at 3σ, SNR 0.5 (knife-edge vs k=0.5)	0.469	exp06
Composite variance ×1.5, T=500	0.82 / 0.87 / 0.91 (SNR 0.1/0.5/2.0)	grid_v1
Ladder ×1.5: raw rung, T=500	1.00 / 0.56 / 0.10 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder ×1.5: ARIMA rung, T=500	0.90 / 0.94 / 0.87 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder ×1.5 t_5 (raw/ARIMA/composite), SNR 0.5	0.43 / 0.74 / 0.16	grid_v4_varbench, grid_v2_misspec
Decision-rule outcome fired	C (raw rung SNR-dependent)	CHANGELOG
Variance ×1.5 across T = 200/500/2000	0.11 / 0.87 / 0.99	grid_v2_T
t_5 collapse and repair (×1.5)	0.16 → 0.75 (tail_cusum)	grid_v2_misspec, grid_v3c
Quieting ×⅔ (only tail_cusum)	0.41 / 0.33	grid_v3c
Persistence-down, best anywhere	0.33 (SNR 2.0)	grid_v1
Multi-break: raw second-event recall	0.00	exp04
Composite level→var second event	0.60 (F1 0.63)	exp04
INDPRO permutation p (composite)	0.008	rd_eval
GFC real-time	data month 2008-09, known 2008-12	rd_realtime
COVID real-time	data 2020-03, ~2 mo before NBER	rd_realtime